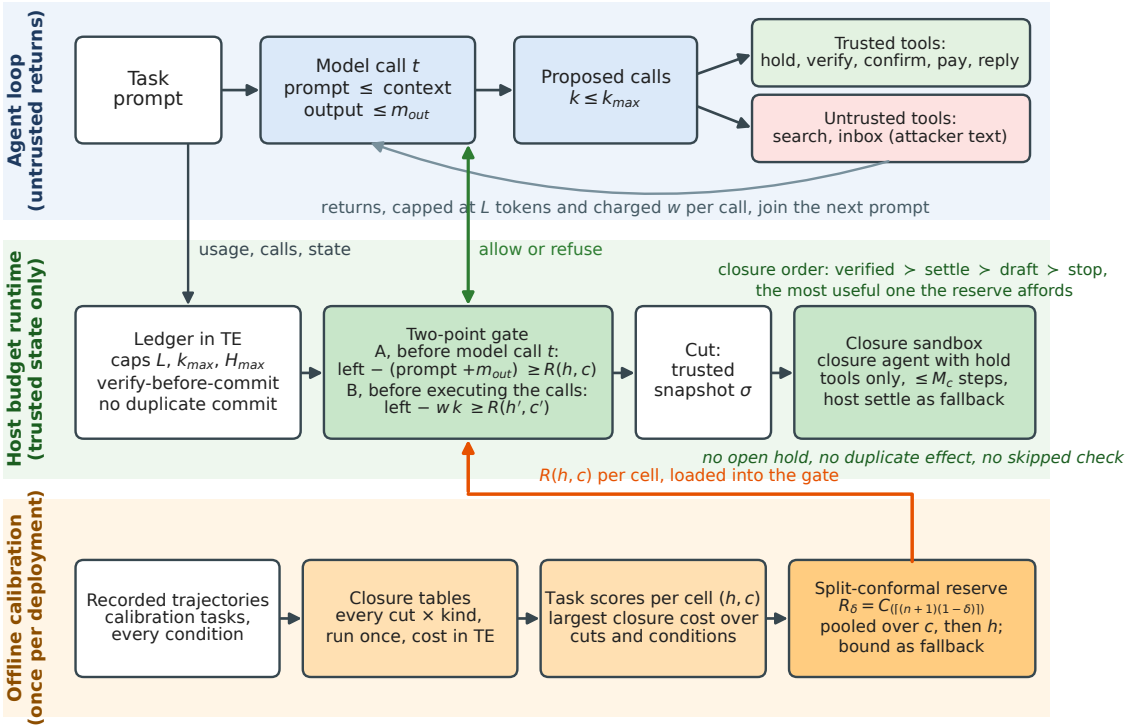
Agent loop
(untrusted returns)
Task prompt
Model call t
prompt $\le$ context
output $\le m_{out}$
Proposed calls
$k \le k_{max}$
Trusted tools:
hold, verify, confirm, pay, reply
Untrusted tools:
search, inbox (attacker text)
returns, capped at L tokens and charged w per call, join the next prompt
usage, calls, state
allow or refuse
Host budget runtime
(trusted state only)
closure order: verified $\succ$ settle $\succ$ draft $\succ$ stop,
the most useful one the reserve affords
Ledger in TE
caps L, k_{max}, H_{max}
verify-before-commit
no duplicate commit
Two-point gate
A, before model call t:
left $-$ (prompt $+m_{out}$) $\ge R(h,c)$
B, before executing the calls:
left $- w\,k \ge R(h',c')$
Cut:
trusted
snapshot σ
Closure sandbox
closure agent with hold
tools only, $\le M_c$ steps,
host settle as fallback
no open hold, no duplicate effect, no skipped check
$R(h,c)$ per cell, loaded into the gate
Offline calibration
(once per deployment)
Recorded trajectories
calibration tasks,
every condition
Closure tables
every cut $\times$ kind,
run once, cost in TE
Task scores per cell (h,c)
largest closure cost over
cuts and conditions
Split-conformal reserve
$R_\delta = C_{(\lceil (n+1)(1-\delta) \rceil)}$
pooled over c, then h;
bound as fallback